

Hippocampal Shape Analysis - Model Evaluation Report

This document summarizes the performance and evaluation of various models across different dataset configurations.

Model Code Descriptions

MLP

The Multi-Layer Perceptron (MLP) model first employs Partial Least Squares Discriminant Analysis (PLS-DA) with 5-fold cross-validation to select the optimal number of components and reduce feature dimensionality. The reduced features are then passed into a standard scikit-learn MLPClassifier for binary classification.

MobileNet

The MobileNet model utilizes a custom PyTorch 1D adaptation of MobileNetV2. It leverages inverted residual blocks and depthwise separable 1D convolutions to efficiently capture sequential/spatial patterns in the features, concluding with a global average pooling layer and linear classifier.

ResNet

The ResNet model is implemented as a custom 1D PyTorch architecture. It features standard residual blocks with skip connections and 1D convolutions, ensuring that gradient vanishing is mitigated while capturing hierarchical features from the hippocampal shape data.

SVM

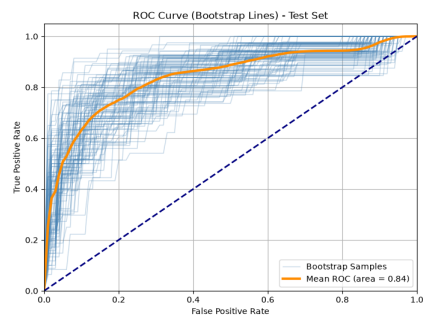
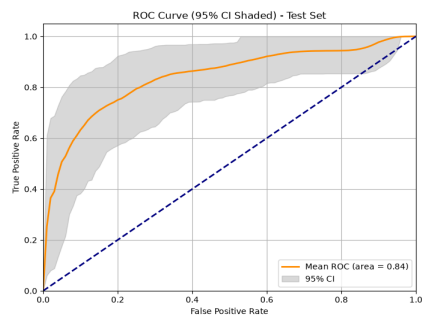
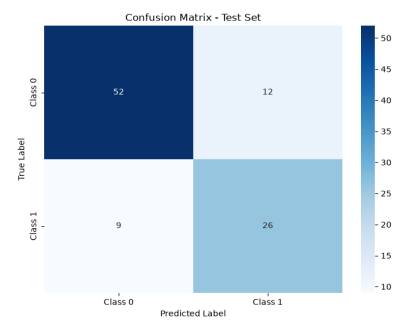
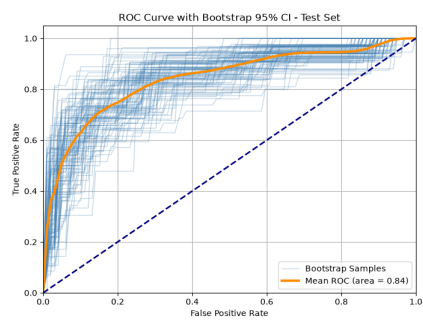
The Support Vector Machine (SVM) pipeline works similarly to the MLP, first applying PLS-DA to find the best component count via cross-validation, and then fitting an SVM classifier (scikit-learn SVC) on the dimension-reduced dataset.

Dataset Configuration: All_Augment_tain

Hemisphere: Left

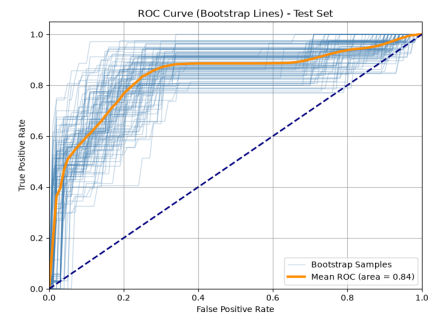
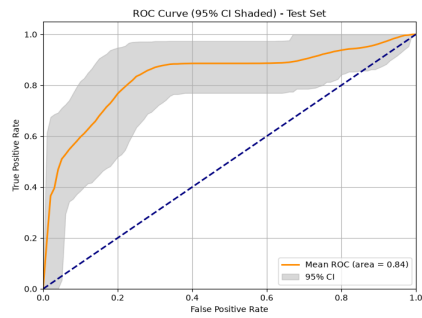
Model: MLP

Metric	Value
Test Accuracy	0.7879
Best PLS Components	10
CV Accuracy (PLS)	0.9899
Class 0 (Precision / Recall / F1)	0.85 / 0.81 / 0.83
Class 1 (Precision / Recall / F1)	0.68 / 0.74 / 0.71



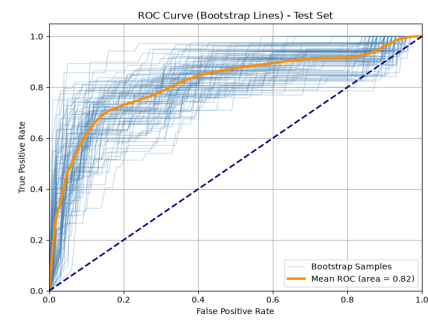
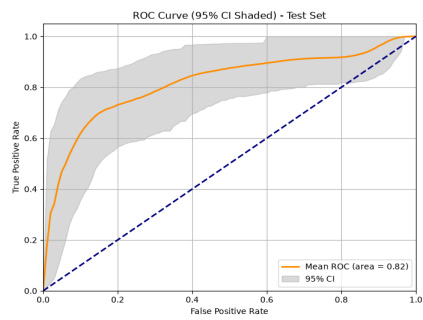
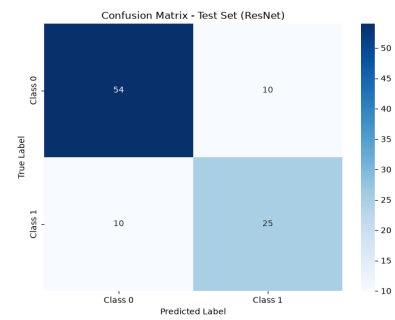
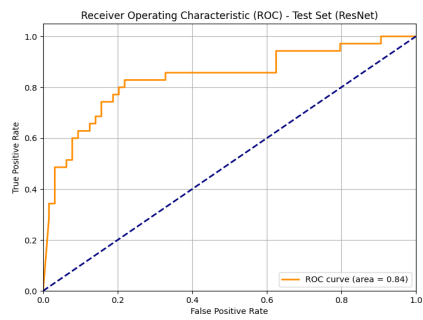
Model: MobileNet

Metric	Value
Test Accuracy	0.8485
Best PLS Components	10
CV Accuracy (PLS)	0.9899
Class 0 (Precision / Recall / F1)	0.87 / 0.91 / 0.89
Class 1 (Precision / Recall / F1)	0.81 / 0.74 / 0.78



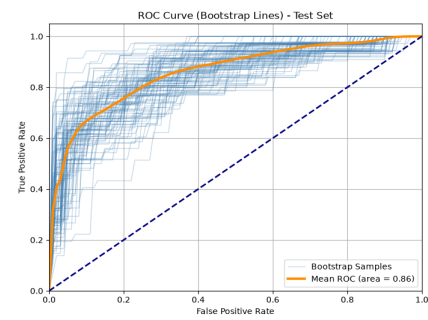
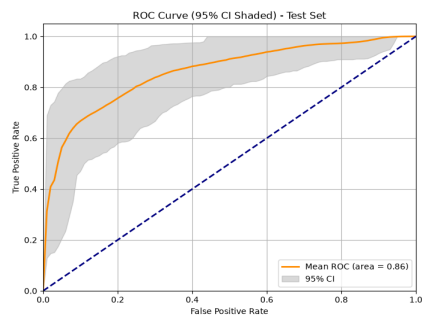
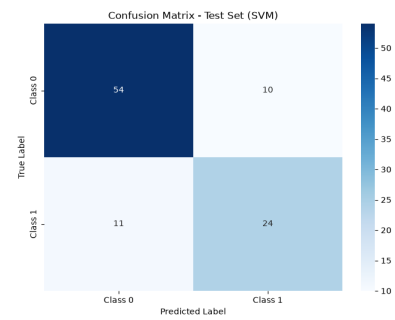
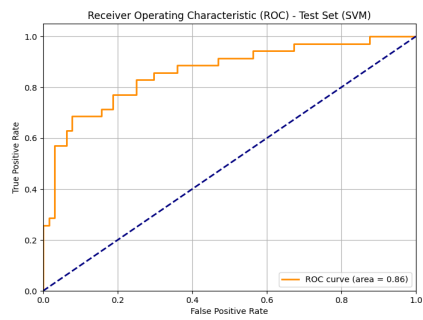
Model: ResNet

Metric	Value
Test Accuracy	0.8081
Best PLS Components	10
CV Accuracy (PLS)	0.9899
Class 0 (Precision / Recall / F1)	0.86 / 0.84 / 0.85
Class 1 (Precision / Recall / F1)	0.72 / 0.74 / 0.73



Model: SVM

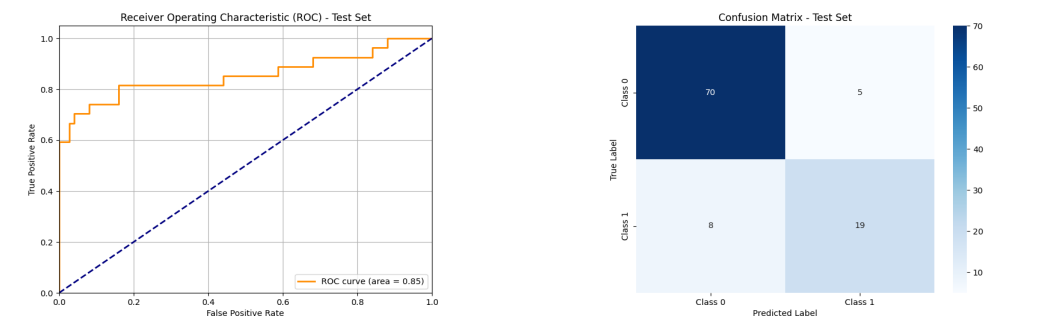
Metric	Value
Test Accuracy	0.7879
Best PLS Components	10
CV Accuracy (PLS)	0.9899
Class 0 (Precision / Recall / F1)	0.83 / 0.84 / 0.84
Class 1 (Precision / Recall / F1)	0.71 / 0.69 / 0.70



Hemisphere: Right

Model: MLP

Metric	Value
Test Accuracy	0.8725
Best PLS Components	5
CV Accuracy (PLS)	0.9983
Class 0 (Precision / Recall / F1)	0.90 / 0.93 / 0.92
Class 1 (Precision / Recall / F1)	0.79 / 0.70 / 0.75

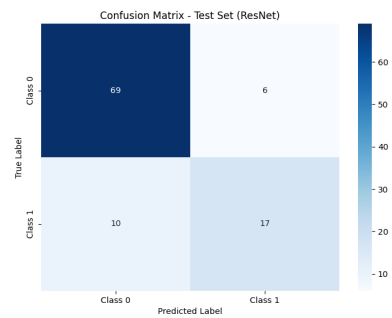
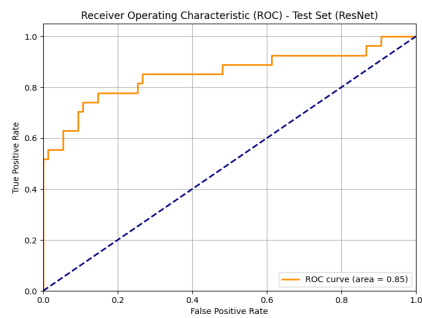


Model: MobileNet

Metric	Value
Test Accuracy	0.8529
Best PLS Components	5
CV Accuracy (PLS)	0.9983
Class 0 (Precision / Recall / F1)	0.87 / 0.95 / 0.90
Class 1 (Precision / Recall / F1)	0.80 / 0.59 / 0.68

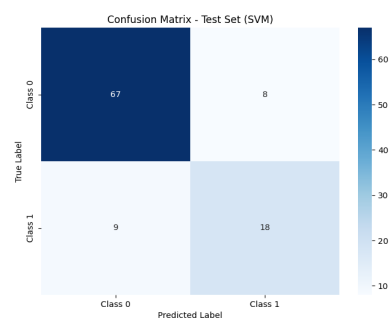
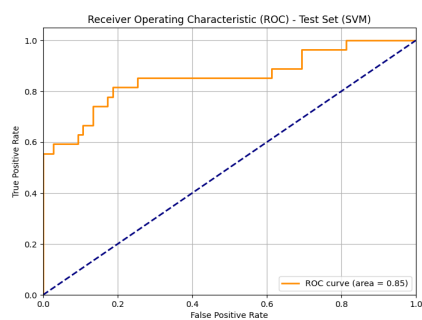
Model: ResNet

Metric	Value
Test Accuracy	0.8431
Best PLS Components	5
CV Accuracy (PLS)	0.9983
Class 0 (Precision / Recall / F1)	0.87 / 0.92 / 0.90
Class 1 (Precision / Recall / F1)	0.74 / 0.63 / 0.68



Model: SVM

Metric	Value
Test Accuracy	0.8333
Best PLS Components	5
CV Accuracy (PLS)	0.9983
Class 0 (Precision / Recall / F1)	0.88 / 0.89 / 0.89
Class 1 (Precision / Recall / F1)	0.69 / 0.67 / 0.68

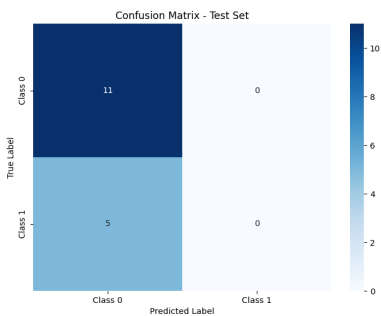
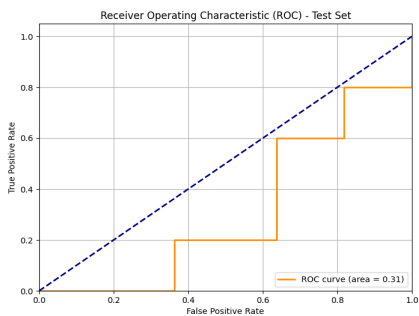


Dataset Configuration: Ds004469

Hemisphere: Left

Model: MLP

Metric	Value
Test Accuracy	0.6875
Best PLS Components	2
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.69 / 1.00 / 0.81
Class 1 (Precision / Recall / F1)	0.69 / 1.00 / 0.81

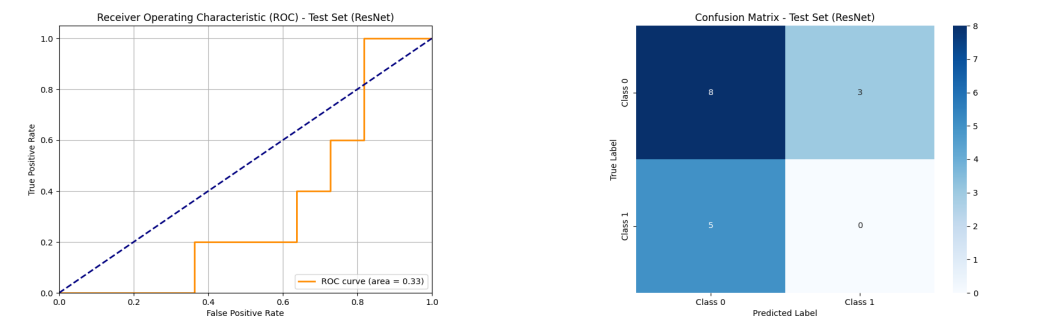


Model: MobileNet

Metric	Value
Test Accuracy	0.2500
Best PLS Components	2
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.44 / 0.36 / 0.40
Class 1 (Precision / Recall / F1)	0.00 / 0.00 / 0.00

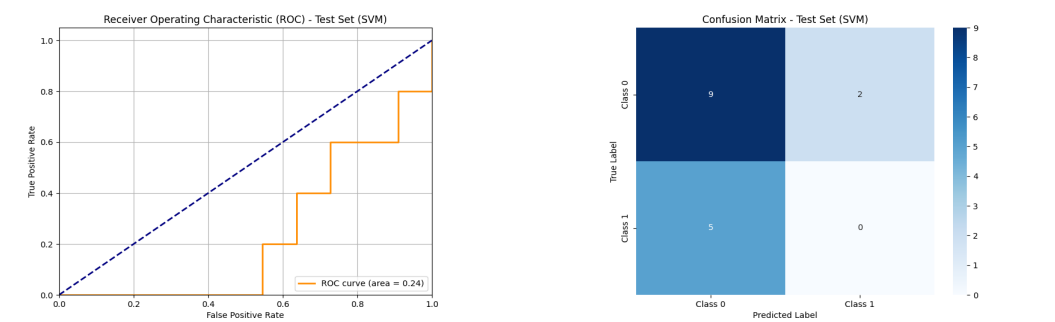
Model: ResNet

Metric	Value
Test Accuracy	0.5000
Best PLS Components	2
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.62 / 0.73 / 0.67
Class 1 (Precision / Recall / F1)	0.00 / 0.00 / 0.00



Model: SVM

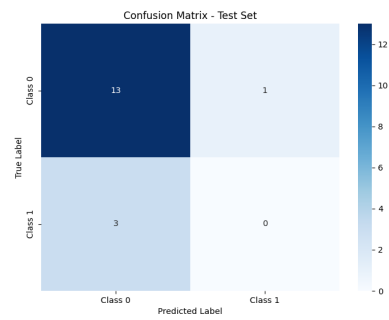
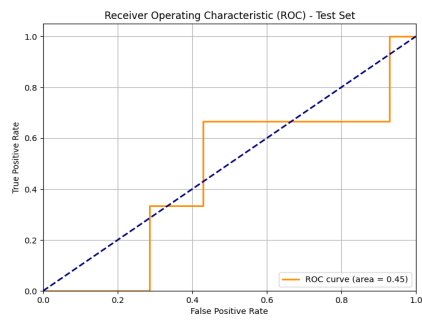
Metric	Value
Test Accuracy	0.5625
Best PLS Components	2
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.64 / 0.82 / 0.72
Class 1 (Precision / Recall / F1)	0.00 / 0.00 / 0.00



Hemisphere: Right

Model: MLP

Metric	Value
Test Accuracy	0.7647
Best PLS Components	5
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.81 / 0.93 / 0.87
Class 1 (Precision / Recall / F1)	0.00 / 0.00 / 0.00

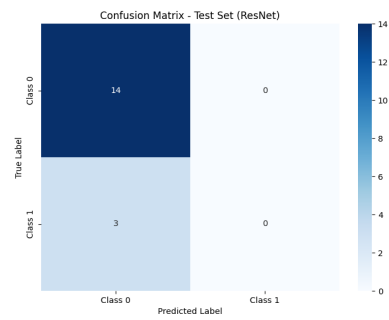
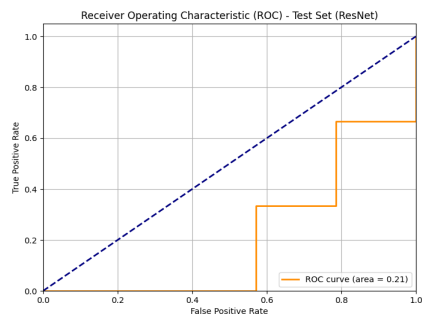


Model: MobileNet

Metric	Value
Test Accuracy	0.8235
Best PLS Components	5
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.82 / 1.00 / 0.90
Class 1 (Precision / Recall / F1)	0.00 / 0.00 / 0.00

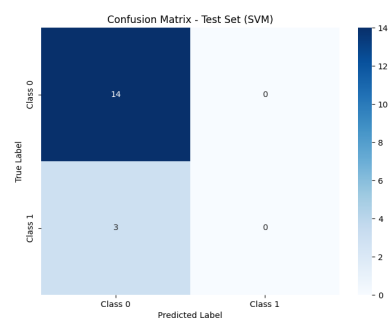
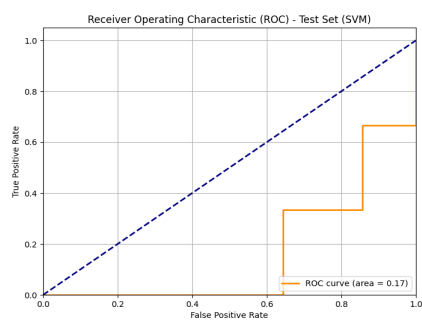
Model: ResNet

Metric	Value
Test Accuracy	0.8235
Best PLS Components	5
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.82 / 1.00 / 0.90
Class 1 (Precision / Recall / F1)	0.00 / 0.00 / 0.00



Model: SVM

Metric	Value
Test Accuracy	0.8235
Best PLS Components	5
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.82 / 1.00 / 0.90
Class 1 (Precision / Recall / F1)	0.00 / 0.00 / 0.00

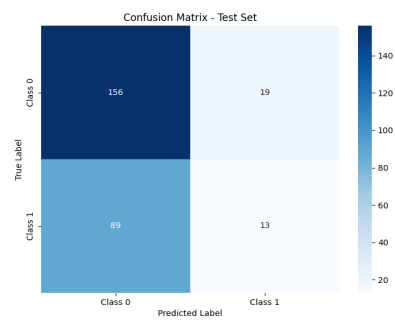
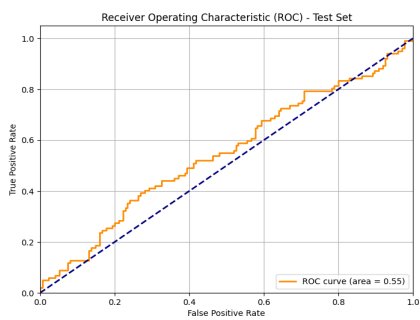


Dataset Configuration: Ds004469Train_Ds005602test

Hemisphere: Left

Model: MLP

Metric	Value
Test Accuracy	0.6101
Best PLS Components	5
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.64 / 0.89 / 0.74
Class 1 (Precision / Recall / F1)	0.41 / 0.13 / 0.19

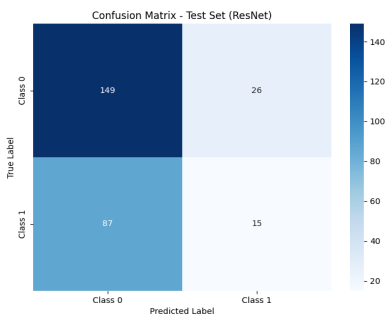
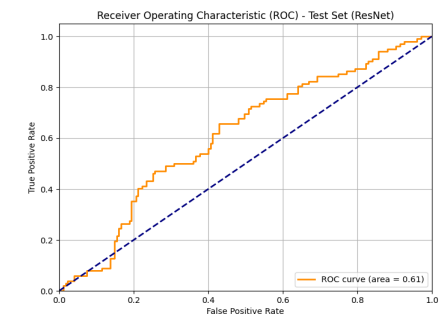


Model: MobileNet

Metric	Value
Test Accuracy	0.5921
Best PLS Components	5
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.63 / 0.87 / 0.73
Class 1 (Precision / Recall / F1)	0.34 / 0.12 / 0.18

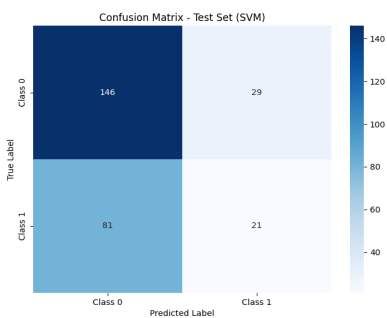
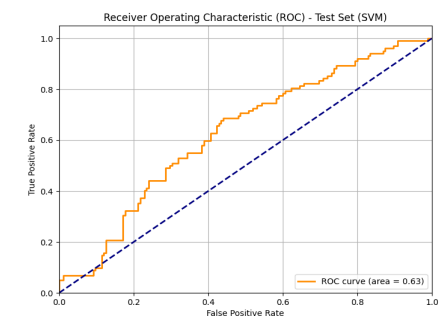
Model: ResNet

Metric	Value
Test Accuracy	0.5921
Best PLS Components	5
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.63 / 0.85 / 0.73
Class 1 (Precision / Recall / F1)	0.37 / 0.15 / 0.21



Model: SVM

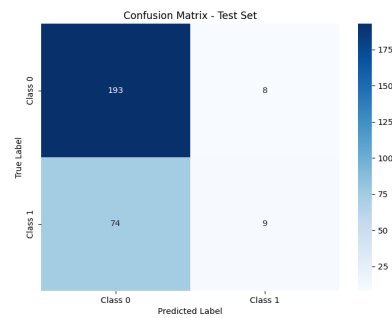
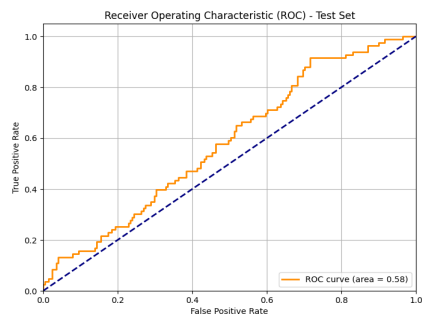
Metric	Value
Test Accuracy	0.6029
Best PLS Components	5
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.64 / 0.83 / 0.73
Class 1 (Precision / Recall / F1)	0.42 / 0.21 / 0.28



Hemisphere: Right

Model: MLP

Metric	Value
Test Accuracy	0.7113
Best PLS Components	2
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.72 / 0.96 / 0.82
Class 1 (Precision / Recall / F1)	0.53 / 0.11 / 0.18

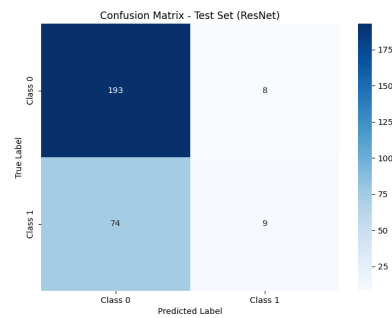
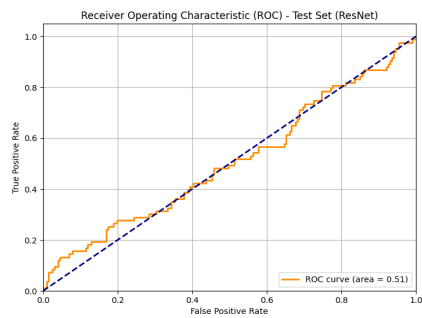


Model: MobileNet

Metric	Value
Test Accuracy	0.7113
Best PLS Components	2
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.72 / 0.96 / 0.82
Class 1 (Precision / Recall / F1)	0.53 / 0.11 / 0.18

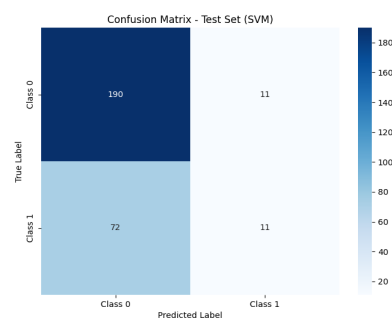
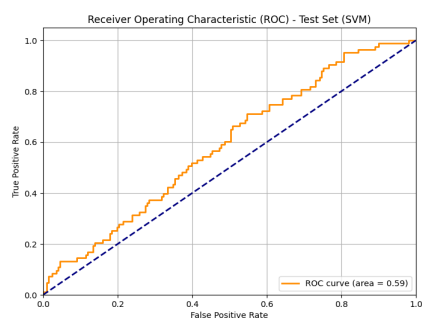
Model: ResNet

Metric	Value
Test Accuracy	0.7113
Best PLS Components	2
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.72 / 0.96 / 0.82
Class 1 (Precision / Recall / F1)	0.53 / 0.11 / 0.18



Model: SVM

Metric	Value
Test Accuracy	0.7077
Best PLS Components	2
CV Accuracy (PLS)	1.0000
Class 0 (Precision / Recall / F1)	0.73 / 0.95 / 0.82
Class 1 (Precision / Recall / F1)	0.50 / 0.13 / 0.21

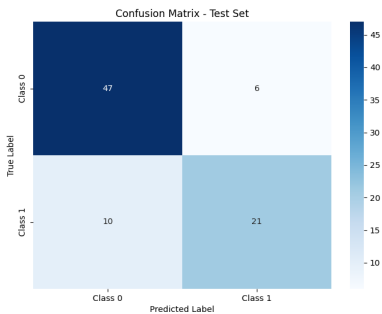
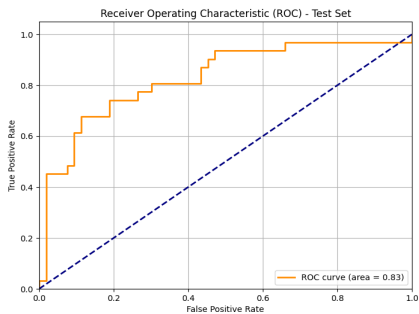


Dataset Configuration: Ds005602

Hemisphere: Left

Model: MLP

Metric	Value
Test Accuracy	0.8095
Best PLS Components	5
CV Accuracy (PLS)	0.9992
Class 0 (Precision / Recall / F1)	0.82 / 0.89 / 0.85
Class 1 (Precision / Recall / F1)	0.78 / 0.68 / 0.72

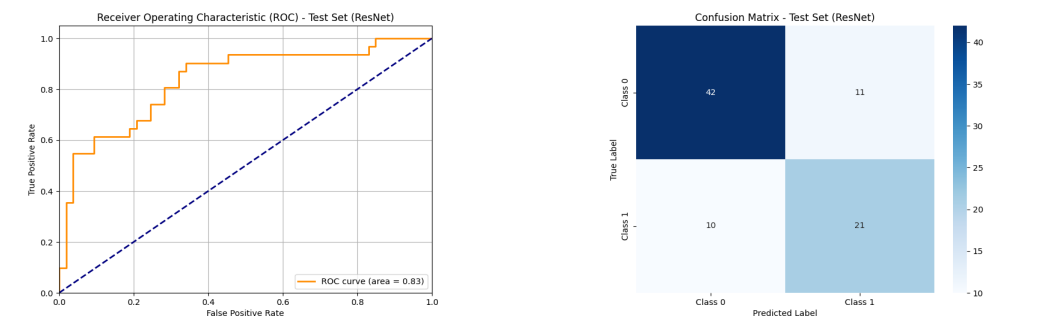


Model: MobileNet

Metric	Value
Test Accuracy	0.7619
Best PLS Components	5
CV Accuracy (PLS)	0.9992
Class 0 (Precision / Recall / F1)	0.80 / 0.83 / 0.81
Class 1 (Precision / Recall / F1)	0.80 / 0.83 / 0.81

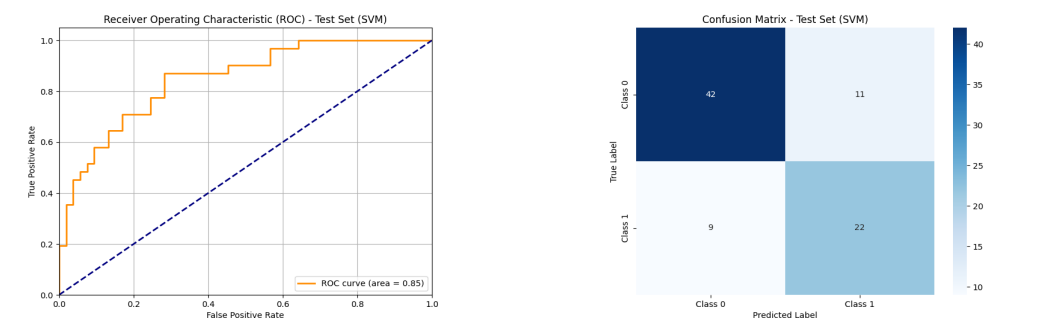
Model: ResNet

Metric	Value
Test Accuracy	0.7500
Best PLS Components	5
CV Accuracy (PLS)	0.9992
Class 0 (Precision / Recall / F1)	0.81 / 0.79 / 0.80
Class 1 (Precision / Recall / F1)	0.66 / 0.68 / 0.67



Model: SVM

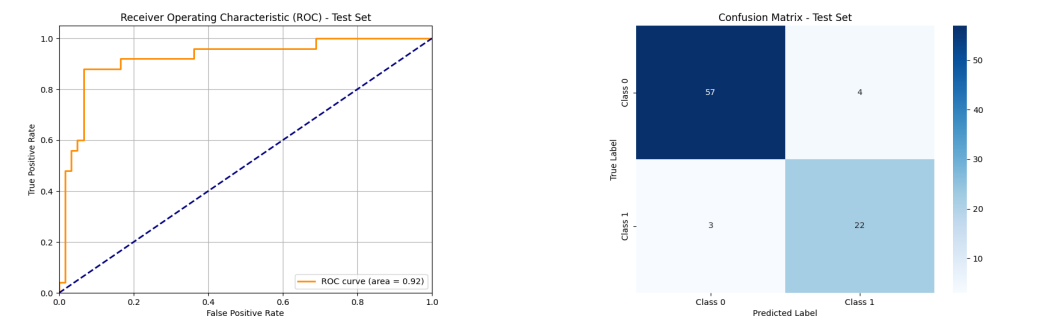
Metric	Value
Test Accuracy	0.7619
Best PLS Components	5
CV Accuracy (PLS)	0.9992
Class 0 (Precision / Recall / F1)	0.82 / 0.79 / 0.81
Class 1 (Precision / Recall / F1)	0.82 / 0.79 / 0.81



Hemisphere: Right

Model: MLP

Metric	Value
Test Accuracy	0.9186
Best PLS Components	5
CV Accuracy (PLS)	0.9986
Class 0 (Precision / Recall / F1)	0.95 / 0.93 / 0.94
Class 1 (Precision / Recall / F1)	0.85 / 0.88 / 0.86

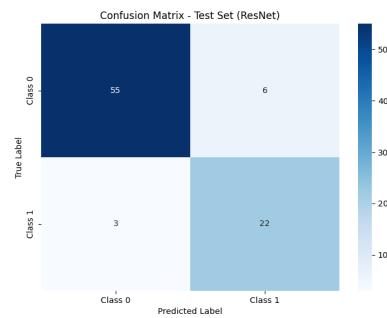
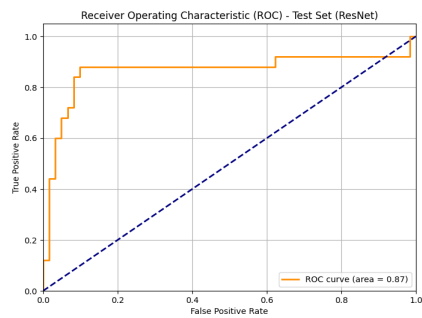


Model: MobileNet

Metric	Value
Test Accuracy	0.8721
Best PLS Components	5
CV Accuracy (PLS)	0.9986
Class 0 (Precision / Recall / F1)	0.93 / 0.89 / 0.91
Class 1 (Precision / Recall / F1)	0.93 / 0.89 / 0.91

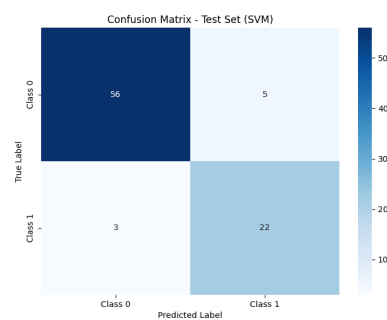
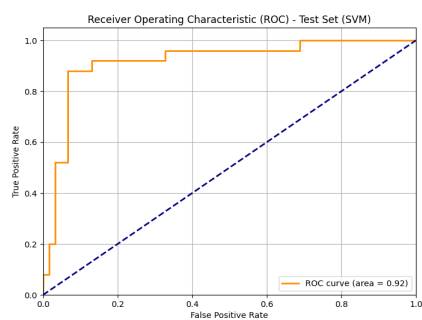
Model: ResNet

Metric	Value
Test Accuracy	0.8953
Best PLS Components	5
CV Accuracy (PLS)	0.9986
Class 0 (Precision / Recall / F1)	0.95 / 0.90 / 0.92
Class 1 (Precision / Recall / F1)	0.79 / 0.88 / 0.83



Model: SVM

Metric	Value
Test Accuracy	0.9070
Best PLS Components	5
CV Accuracy (PLS)	0.9986
Class 0 (Precision / Recall / F1)	0.95 / 0.92 / 0.93
Class 1 (Precision / Recall / F1)	0.81 / 0.88 / 0.85

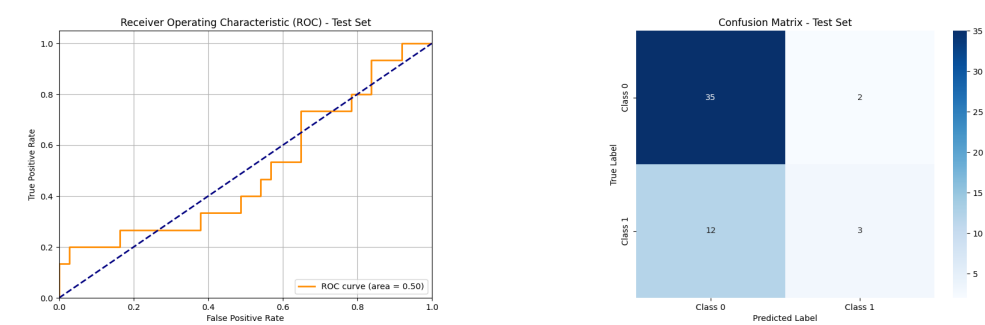


Dataset Configuration: Ds005602Train_Ds004469test

Hemisphere: Left

Model: MLP

Metric	Value
Test Accuracy	0.7308
Best PLS Components	10
CV Accuracy (PLS)	0.9914
Class 0 (Precision / Recall / F1)	0.74 / 0.95 / 0.83
Class 1 (Precision / Recall / F1)	0.60 / 0.20 / 0.30

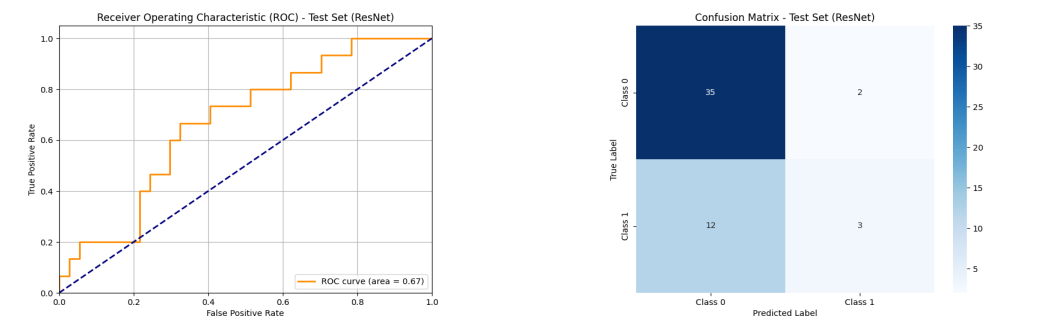


Model: MobileNet

Metric	Value
Test Accuracy	0.7308
Best PLS Components	10
CV Accuracy (PLS)	0.9914
Class 0 (Precision / Recall / F1)	0.74 / 0.95 / 0.83
Class 1 (Precision / Recall / F1)	0.60 / 0.20 / 0.30

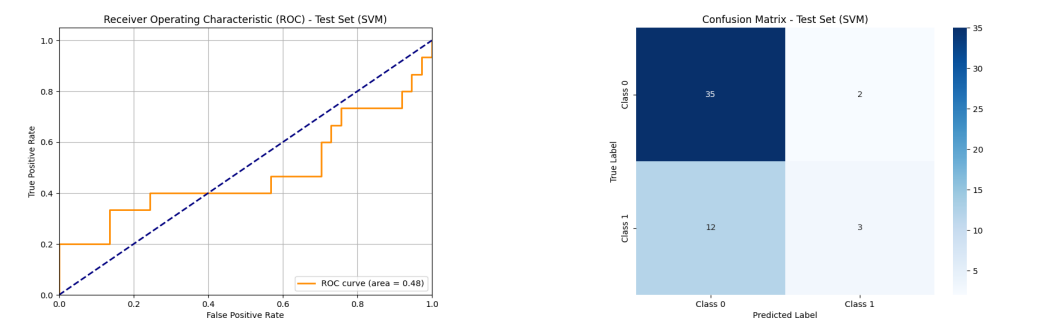
Model: ResNet

Metric	Value
Test Accuracy	0.7308
Best PLS Components	10
CV Accuracy (PLS)	0.9914
Class 0 (Precision / Recall / F1)	0.74 / 0.95 / 0.83
Class 1 (Precision / Recall / F1)	0.60 / 0.20 / 0.30



Model: SVM

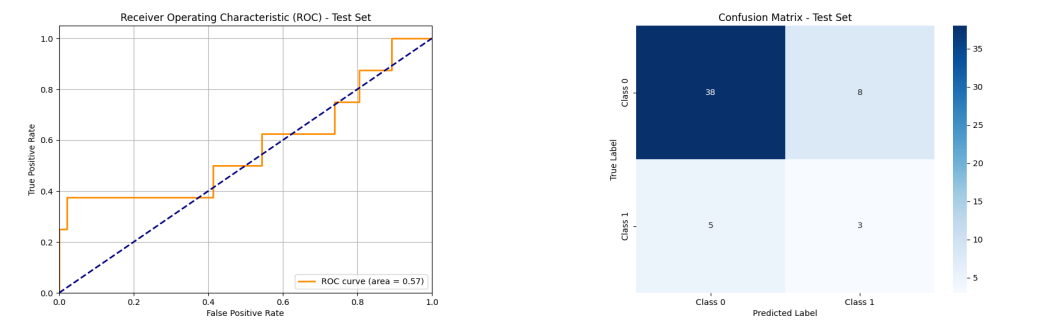
Metric	Value
Test Accuracy	0.7308
Best PLS Components	10
CV Accuracy (PLS)	0.9914
Class 0 (Precision / Recall / F1)	0.74 / 0.95 / 0.83
Class 1 (Precision / Recall / F1)	0.60 / 0.20 / 0.30



Hemisphere: Right

Model: MLP

Metric	Value
Test Accuracy	0.7593
Best PLS Components	5
CV Accuracy (PLS)	0.9965
Class 0 (Precision / Recall / F1)	0.88 / 0.83 / 0.85
Class 1 (Precision / Recall / F1)	0.27 / 0.38 / 0.32

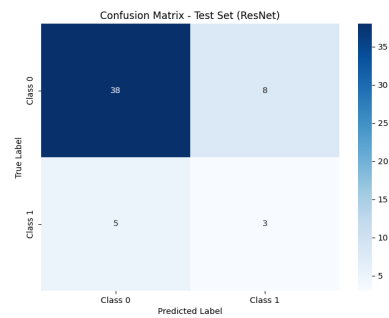
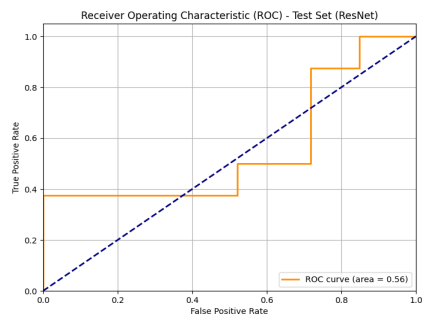


Model: MobileNet

Metric	Value
Test Accuracy	0.7778
Best PLS Components	5
CV Accuracy (PLS)	0.9965
Class 0 (Precision / Recall / F1)	0.89 / 0.85 / 0.87
Class 1 (Precision / Recall / F1)	0.30 / 0.38 / 0.33

Model: ResNet

Metric	Value
Test Accuracy	0.7593
Best PLS Components	5
CV Accuracy (PLS)	0.9965
Class 0 (Precision / Recall / F1)	0.88 / 0.83 / 0.85
Class 1 (Precision / Recall / F1)	0.27 / 0.38 / 0.32



Model: SVM

Metric	Value
Test Accuracy	0.7778
Best PLS Components	5
CV Accuracy (PLS)	0.9965
Class 0 (Precision / Recall / F1)	0.89 / 0.85 / 0.87
Class 1 (Precision / Recall / F1)	0.30 / 0.38 / 0.33

